SOFTWARE ENGINEER INTERVIEW

# Software Engineering & System Simulation

Production Software Development · GPU LLM Serving Research

**Yu-Jui Liang**

# Education & Professional Experience

## Education

**National Taiwan University**
M.S. Candidate, Data Science Master's Degree Program
Expected Graduation: June 2027

**National Chengchi University**
B.S. in Computer Science
Double Major in Electronic Physics
Sep. 2021 – Jun. 2025

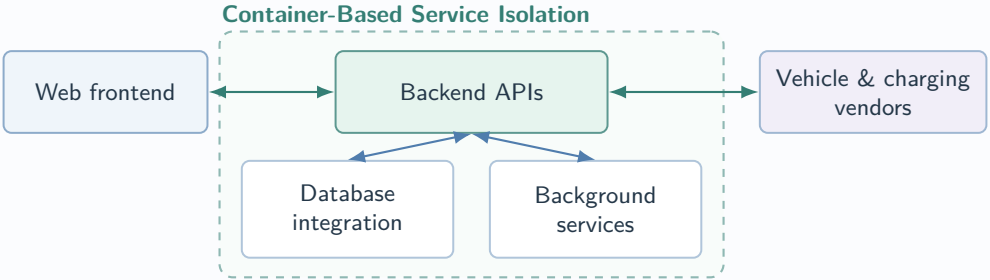
## Professional Experience

**AIMobile Co., Ltd.**
Inventec Group
Contract Software Engineer
Apr. 2025 – Jul. 2025

**Neopower Technologies Co., Ltd.**
Contract Software Engineer
Jun. 2023 – Mar. 2025

# NeoPower: Backend System Architecture

## Electric Bus Charging Management Platform



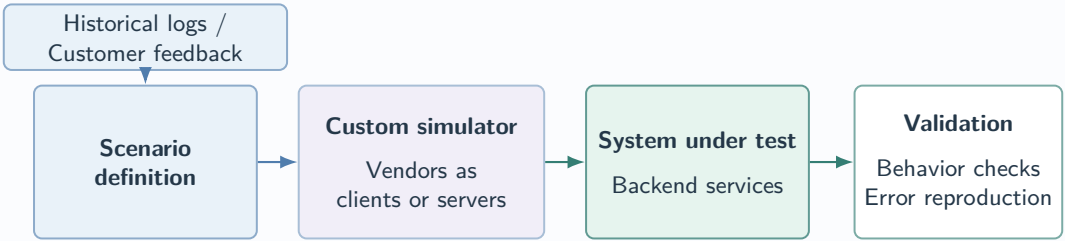

**My contributions**

System planning & backend development / Database integration

Log analysis & debugging / Deployment & maintenance

# NeoPower: Scenario-Based System Simulation

**I developed a custom simulator for vendor interaction and failure reproduction.**




**High Load**

Many buses arrive at once

**Network Failure**

Mass disconnection of stations / chargers

# NeoPower: Testing & Production Reliability

## Pre-Deployment Testing

**Simulator-based validation**

Normal / abnormal conditions
Regression testing

Tests pass

Deploy updated system

## Post-Deployment Validation

**Charging-power allocation algorithms**

Cross-Site Comparison: similar operating conditions

Some sites
**Previous algorithm**

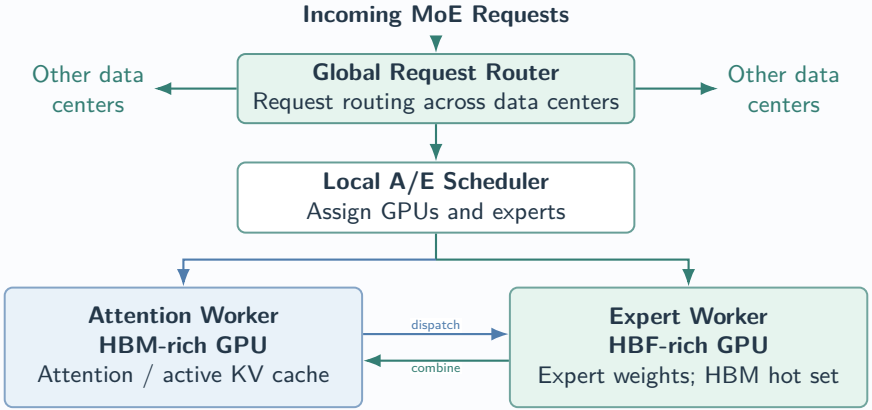
Other sites
**Updated algorithm**

**Compare operating behavior**

System logs / operating data
Customer feedback

# MoE Serving: HBM/HBF-Aware Attention–Expert Disaggregation

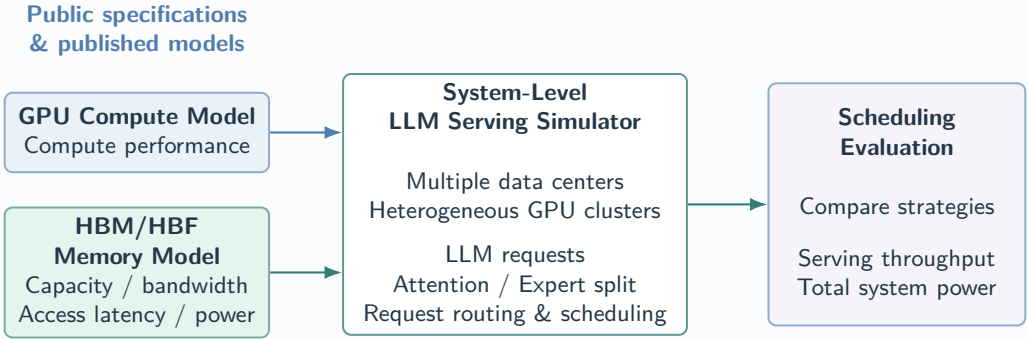
**Co-design worker memory profiles and A/E placement to improve SLO goodput per watt.**




**Goal:** Balance trade-offs among latency, power, and throughput.

# Research Methodology: System-Level Simulation

**Model heterogeneous hardware to evaluate LLM serving scheduling.**




**Architectural starting point:** open-source GPU-cluster serving simulator

**Planned integration:** published GPU-memory and storage-system models

# Thank You

Questions & Discussion